

FEATURE SCALING

AIM

To understand the importance of feature scaling and to apply standardization and normalization (min-max scaling) on a data set.

PROGRAM

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler, MinMaxScaler

np.set_printoptions(precision=4, suppress=True, linewidth=80)

df = pd.read_csv('pre-process_datasample.csv')
print(df)

df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

# Handling missing numerical values
age = SimpleImputer(strategy="mean", missing_values=np.nan)
Salary = SimpleImputer(strategy="mean", missing_values=np.nan)
age.fit(features[:, [1]])
Salary.fit(features[:, [2]])
features[:, [1]] = age.transform(features[:, [1]])
features[:, [2]] = Salary.transform(features[:, [2]])
print(features)

# One-hot encoding of the Country column
oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, [0]])
print(Country)

final_set = np.concatenate((Country, features[:, [1, 2]]), axis=1).astype(float)
print(final_set)

# Standardization
sc = StandardScaler()
sc.fit(final_set)
feat_standard_scaler = sc.transform(final_set)
print("\nStandard Scaler output:\n", feat_standard_scaler)
print("Mean of each column :", feat_standard_scaler.mean(axis=0))
print("Std of each column :", feat_standard_scaler.std(axis=0))

# Normalization (Min-Max scaling)
mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)
feat_minmax_scaler = mms.transform(final_set)
print("\nMin-Max Scaler output:\n", feat_minmax_scaler)
print("Min of each column :", feat_minmax_scaler.min(axis=0))
print("Max of each column :", feat_minmax_scaler.max(axis=0))
```

OUTPUT

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No

```

3   Spain  38.0  61000.0      No
4   Germany 40.0      NaN      Yes
5   France  35.0  58000.0      Yes
6   Spain   NaN  52000.0      No
7   France  48.0  79000.0      Yes
8     NaN   50.0  83000.0      No
9   France  37.0  67000.0      Yes
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 63777.77777777778]
 ['France' 35.0 58000.0]
 ['Spain' 38.77777777777778 52000.0]
 ['France' 48.0 79000.0]
 ['France' 50.0 83000.0]
 ['France' 37.0 67000.0]]
[[1. 0. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [1. 0. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [1. 0. 0.]
 [1. 0. 0.]]
[[ 1.      0.      0.      44.      72000. ]
 [ 0.      0.      1.      27.      48000. ]
 [ 0.      1.      0.      30.      54000. ]
 [ 0.      0.      1.      38.      61000. ]
 [ 0.      1.      0.      40.      63777.7778]
 [ 1.      0.      0.      35.      58000. ]
 [ 0.      0.      1.      38.7778 52000. ]
 [ 1.      0.      0.      48.      79000. ]
 [ 1.      0.      0.      50.      83000. ]
 [ 1.      0.      0.      37.      67000. ]]

```

Standard Scaler output:

```

[[ 1.      -0.5      -0.6547  0.7589  0.7495]
 [-1.      -0.5      1.5275 -1.7115 -1.4382]
 [-1.       2.      -0.6547 -1.2756 -0.8913]
 [-1.      -0.5      1.5275 -0.113  -0.2532]
 [-1.       2.      -0.6547  0.1776  0.      ]
 [ 1.      -0.5      -0.6547 -0.549  -0.5267]
 [-1.      -0.5      1.5275  0.      -1.0736]
 [ 1.      -0.5      -0.6547  1.3401  1.3875]
 [ 1.      -0.5      -0.6547  1.6308  1.7521]
 [ 1.      -0.5      -0.6547 -0.2583  0.2937]]

```

Mean of each column : [0. -0. -0. -0. 0.]

Std of each column : [1. 1. 1. 1. 1.]

Min-Max Scaler output:

```

[[1.      0.      0.      0.7391  0.6857]
 [0.      0.      1.      0.      0.      ]
 [0.      1.      0.      0.1304  0.1714]
 [0.      0.      1.      0.4783  0.3714]
 [0.      1.      0.      0.5652  0.4508]
 [1.      0.      0.      0.3478  0.2857]
 [0.      0.      1.      0.5121  0.1143]
 [1.      0.      0.      0.913   0.8857]
 [1.      0.      0.      1.      1.      ]
 [1.      0.      0.      0.4348  0.5429]]

```

Min of each column : [0. 0. 0. 0. 0.]

Max of each column : [1. 1. 1. 1. 1.]

RESULT

Thus feature scaling was performed on the data set. StandardScaler produced columns with zero mean and unit variance, and MinMaxScaler brought every column into the range 0 to 1.